**MAKING THE RIGHT CONNECTIONS: IMPROVING THE COMPREHENSION OF STRUGGLING READERS**

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## Executive Summary

### Project Goals

The major goals of this project are to produce one or more fully developed questioning intervention(s) aimed at improving the reading comprehension of students at risk or identified as having disabilities. Intended products include:

(a) one or more prototypes that are feasible and promising, with manuals;
(b) one or more observational checklists to be used to document fidelity; and
(c) training materials to prepare school-based personnel for implementation.

We also aim to identify for whom (subgroups of struggling readers) and under what conditions (type and timing of questions, intensity of intervention) the intervention is most promising.

### Extent to Which Goals have been Met

In Year 3, we completed or are in the process of completing the following objectives to meet the above project goals, as described below:

**(a) Finalize core intervention questions, materials, and procedures to be implemented in the Year 3 pilot study (complete).** We have met this objective by revising our intervention package called "In-the-Text Connections." The intervention package includes:

- Six narrative and six informational passages, each with 5 questions designed to help students make important connections while they read.
- Teacher and student materials to implement the intervention, including teacher scripts, vocabulary cards, question cards with feedback, bookmarks, teacher and student versions of the passages in folders, score cards, and incentives.
- A fully developed set of procedures and fidelity tools for implementing the intervention.

**(b) Develop materials and procedures to train school-based personnel to implement intervention (complete).** We have created a full-day teacher workshop that includes:

- A power point covering intervention purpose, theoretical framework, overview of materials and procedures, demonstrations and illustrations (including video).
- Procedures for scaffolding teachers' practice in using the intervention materials and implementing the procedures.

**(c) Recruit teachers to implement the intervention (complete)**. Five teachers from four schools agreed to implement the intervention. In three schools where teachers were not available to implement the intervention, research staff (tutors; $n = 4$) implemented the intervention. This configuration will allow us to conduct a quasi-experimental comparison of the effects of teachers vs. tutors on student outcomes.

**(d) Train the teachers to implement the intervention (complete).** Five teachers and four tutors attended the teacher workshop and implemented the intervention for six weeks.

**(e) Identify struggling readers (complete).** We have met this objective by conducting a systematic screening process to identify 59 struggling readers to participate in the intervention. Specifically, consented fourth-grade students ($n$ = 460) were screened in two phases. In Screening Phase I, students whose average scores from two Curriculum-Based Measurement (CBM) Maze probes were at or below the 25th percentile were identified as eligible for Screening Phase II. Those students identified as eligible for Screening Phase II were individually administered CBM oral reading probes to eliminate students whose primary reading difficulties were decoding/fluency. Those who scored greater than 90 wrc in one min (and who were confirmed to be struggling readers based on being at or below the 50th percentile on their district MAP test) were identified as eligible for the intervention ($n$ = 59).

**(f) Conduct pilot study (complete).** We have met this objective by conducting a two trial runs and a pilot study to determine feasibility and promise of the intervention on students' comprehension of texts of stories and biographies.

**(g) Analyze and interpret data from pilot study to examine feasibility and student outcomes (in process)**. Preliminary analyses are complete or in progress. General findings indicate that:

- Teachers and tutors implemented the intervention with high levels of fidelity.
- Teachers and tutors rated the intervention as feasible to implement.
- Thus far, students in the intervention group appear to have benefitted from the intervention in terms of the numbers of questions answered correctly with prompting, but only in the "stories" phase of the intervention. Our primary dependent variables (related to verbal recalls) are still being coded, and analyses will be completed in Fall of 2014.

**(h) Finalize the intervention based on results of Year 3 study activities (in process).** When data analyses are complete, we will make final revisions to all intervention materials, fidelity tools, and teacher training materials.

**Contributions**

While we are still in process of analyzing data in order to address the primary goals of the project, Year 3 findings have some potentially important contributions to research. We have developed an intervention prototype that is based on a strong theoretical framework, and that teachers view as feasible and acceptable. Evidence from Year 3 will shed light on the promise of the intervention for subgroups of struggling readers and for different text types (stories and informational texts).